19

Electromagnetics I Final Exam.

Class: 電機三 Student ID.: 411087019 Name: 林書宇

(12/27/2023)

Note: (1)考試時間 100 分鐘;(2)不可使用任何電子設備;(3)請遵守考試規則;(4)直接在本試題紙上作答;(5)請將答題內容乾淨工整

A. Selections (50%) 下列各題請選擇最正確的答案，直接填入題前空格中。

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( ) Given a straight piece of homogeneous ohmic material of conductivity σ, length ℓ, and uniform cross section area S. What is its resistance? [1.] $\sigma\ell/S$; [2.] $\ell/(\sigma S)$; [3.] $\sigma S/\ell$; [4.] $S/(\sigma\ell)$; [5.] none. (Ω)

( ) Suppose D_{in} and D_{it}, $i = 0,1$ denote the normal and tangential components, respectively, of electric flux density in medium i; and E_{in} and E_{it}, $i = 0,1$ denote the normal and tangential components, respectively, of electric field intensity in medium i. Also, ρ_s denotes the surface free charge density at the interface of medium 1 and 2. Which of the following about the boundary conditions of the electrostatic field at the interface between two different perfect dielectrics is wrong? [1.] $D_{1n} - D_{2n} = \rho_s$ [2.] $D_{1t} = D_{2t}$ under any circumstance; [3.] $E_{1t} = E_{2t}$; [4.] $D_{1n} = D_{2n}$, if there is no free charge available at the interface of the two media; [5.] none.

3. ( ) With **D**, **E**, and **P** denote the electric flux density, electric field intensity, and polarization vector, respectively; which of the following expressions about dielectrics in an electrostatic field is wrong? [1.] $\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P}$; [2.] Induced volume bound-charge density $\rho_p = -\nabla\cdot\mathbf{P}$; [3.] Induced surface bound-charge density $\rho_{ps} = \mathbf{P}\cdot\mathbf{a}_n$, where $\mathbf{a}_n$ is the outward normal vector to the surface bounding the dielectric material; [4.] Polarization vector $\mathbf{P} = \varepsilon_0\epsilon_r\mathbf{E}$ for simple media, where ϵ_r is the relative permittivity of the dielectric; [5.] none.

4. ( ) A parallel plate capacitor with two lossy media filled between the plates as shown in Fig. 1. Given σ_1, σ_2, ϵ_1, ϵ_2, $\mathcal{V}$, the area of the plates S and the thickness of the lossy media d_1 and d_2. What is the magnitude of the current density $\mathbf{J}_1$?

[1.] $\frac{\sigma_1\sigma_2\mathcal{V}}{\sigma_1 d_2+\sigma_2 d_1}$, [2.] $\frac{\sigma_1\sigma_2\mathcal{V}}{\sigma_1 d_1+\sigma_2 d_2}$, [3.] $\frac{\sigma_1\mathcal{V}}{\sigma_2 d_1}$, [4.] $\frac{\sigma_2\mathcal{V}}{\sigma_1 d_1}$, [5.] none.

5. ( ) Same as the previous problem, what is the charge density ρ_{sl} (C/m^2) at the interface of two dielectrics?

[1.] $\frac{(\epsilon_1\sigma_1-\epsilon_2\sigma_2)\mathcal{V}}{\sigma_2 d_2+\sigma_1 d_1}$, [2.] $\frac{(\epsilon_2\sigma_1-\epsilon_1\sigma_2)\mathcal{V}}{\sigma_2 d_2+\sigma_1 d_1}$, [3.] $\frac{(\epsilon_2\sigma_1-\epsilon_1\sigma_2)\mathcal{V}}{\sigma_2 d_1+\sigma_1 d_2}$, [4.] $\frac{(\epsilon_1\sigma_1-\epsilon_2\sigma_2)\mathcal{V}}{\sigma_2 d_1+\sigma_1 d_2}$, [5.] none.

6. ( ) The electrostatic energy stored in the region of space $R > b$ around an electric dipole of moment **p** is [1.] $\frac{p^2}{16\pi\epsilon_0 b^3}$; [2.] $\frac{p^2}{48\pi\epsilon_0 b^3}$; [3.] $\frac{p^2}{32\pi\epsilon_0 b^3}$; [4.] $\frac{p^2}{12\pi\epsilon_0 b^3}$; [5.] none.

$W = \frac{1}{2}\int \varepsilon E^2$ $4\pi R^2 E = \frac{Q}{\varepsilon}$

7. ( ) Assume that the $z = 0$ plane separates two lossless dielectric regions with $\epsilon_{r1} = 2$ and $\epsilon_{r2} = 3$. If we know that $\mathbf{E}_1$ in region 1 ($z > 0$) is $\mathbf{E}_1 = \mathbf{a}_x 2y + \mathbf{a}_y(2x - y) + \mathbf{a}_z(5 + z)$, what is the $\mathbf{a}_y$ component of $\mathbf{D}_2$ in region 2 at the interface of the two dielectrics (at $z = 0$)? [1.] $\epsilon_0(10/3)$; [2.] $2y$; [3.] $-3x\epsilon_0$; [4.] $(6x - 3y)\epsilon_0$; [5.] none.

8. ( ). A point charge $+Q$ is placed at $(x, y, z) = (0, d, 0)$, and a grounded conducting plate with infinite extension is placed at $y = 0$. Then the electric potential $V(0, d/2, 0) =$ [1.] $\frac{Q}{3\pi\varepsilon_0 d}$ [2.] $\frac{Q}{4\pi\varepsilon_0 d}$ [3.] $\frac{3Q}{2\pi\varepsilon_0 d}$ [4.] 0 [5.] none.

9. ( ) Same as the previous problem, the surface charge density ρ_s on the surface of the conducting plate can be found as $\rho_s =$

[1.] $-\frac{Qd}{4\pi(x^2+d^2+z^2)^{\frac{3}{2}}}$ [2.] $-\frac{Qd}{2\pi(x^2+d^2+z^2)^{\frac{1}{2}}}$ [3.] $-\frac{Qd}{2\pi(x^2+d^2+z^2)^{\frac{3}{2}}}$ [4.] $-Q$ [5.] none.

10. ( ) The polarization vector in a dielectric cube of side L centered at the origin is given by $\mathbf{P} = P_0(\mathbf{a}_x 2x + \mathbf{a}_y 2y + \mathbf{a}_z 2z)$. What is the total volume bound-charge (not density!) ? [1.] $-8P_0L^3$; [2.] $-6P_0L^3$; [3.] $3P_0L^3$; [4.] $6P_0L^3$; [5.] none.

Fig. 1 Fig. 2

B. Calculations (50%) 請於本試題紙正、反面空白處標明題號作答，除第 1 題(b)小題外，所有題目均須有完整計算或說明過程，只寫答案不予計分。

1. (10%) Please answer the following questions.
(a) Derive the boundary conditions for the tangential and normal components of the polarization vector **P** at an interface between two perfect dielectric media with dielectric constants ϵ_{r1} and ϵ_{r2}.
(b) Describe the boundary conditions for the tangential and normal components of the conduction current density **J** at an interface between two conductors with conductivities σ_1 and σ_2.

2. (15%) A parallel-plate capacitor of width w, length L, and separation d is partially filled with a dielectric medium of dielectric constant ϵ_r, as shown in Fig. 2. A battery of V_0 volts is connected between the plates. (a) Find **D**, **E**, and ρ_s in each region. (b)Find the energy stored in each region. (c)Find distance x such that the electrostatic energy stored in each region is the same.

3. (10%) A d-c voltage V_0 is applied across a cylindrical capacitor of length L. The radii of the inner and outer conductors are a and b, respectively. The space between the conductors is filled with two different lossy dielectrics having respectively, permittivity ϵ_1 and conductivity σ_1 in the region $a < r < c$ and permittivity ϵ_2 and conductivity σ_2 in the region $c < r < b$. Determine (a) the current density in each region. (b)The surface charge densities on the inner and outer conductors and at the interface between the two dielectrics.

4. (15%) A parallel-plate capacitor has width w (in x-direction), depth L (in z-direction), and separation d (in y-direction). The two conducting plates are located at $y = 0$ and $y = d$, and the space between them is filled with a dielectric whose permittivity varies linearly from ϵ_1 at the left end of both plates ($x = 0$) to ϵ_2 at the right end of both plates ($x = w$). A battery of V_0 volts is connected between the plates. Neglect fringing effect. (a) Find **D**, **E** in the dielectric. (b) Find the surface charge density ρ_s on the top conducting plate. (c) Find the capacitance.

1.

(a)

2 $P = D - \varepsilon_0 E = (\varepsilon_r - \varepsilon_0)E$

(b)

$J = \sigma E$

Class:_____ Student ID.: 411087002 Name: [illegible]

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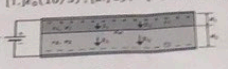

Fig. 1

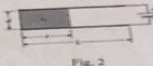

Fig. 2

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+8

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+14

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+7

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+8

1. (a) $\vec{P}=\vec{D}-\varepsilon_0\vec{E}$

$\boxed{8}$

$=\varepsilon_0\varepsilon_r\vec{E}-\varepsilon_0\vec{E}$

$\vec{E}=\frac{\vec{P}}{\varepsilon_0(\varepsilon_r-1)}$

$\vec{E}_{1t}=\vec{E}_{2t} \Rightarrow \frac{\vec{P}_{1t}}{\varepsilon_0(\varepsilon_{r1}-1)}=\frac{\vec{P}_{2t}}{\varepsilon_0(\varepsilon_{r2}-1)}$

$\Rightarrow \frac{\vec{P}_{1t}}{\varepsilon_{r1}-1}=\frac{\vec{P}_{2t}}{\varepsilon_{r2}-1}$ #

4

$\varepsilon_0\varepsilon_{r1}\vec{E}_{1n}=\varepsilon_0\varepsilon_{r2}\vec{E}_{1n} \Rightarrow \frac{\varepsilon_{r1}\vec{P}_{2n}}{\varepsilon_{r1}-1}=\frac{\varepsilon_{r2}\vec{P}_{2n}}{\varepsilon_{r2}-1}$

b) $\frac{J_{1t}}{J_{2t}}=\frac{\varepsilon_{r1}}{\varepsilon_{r2}} \Rightarrow \varepsilon_{r2}\vec{J}_{1t}=\varepsilon_{r1}\vec{J}_{2t}$

$\vec{J}_{1n}=\vec{J}_{2n}$ #

4